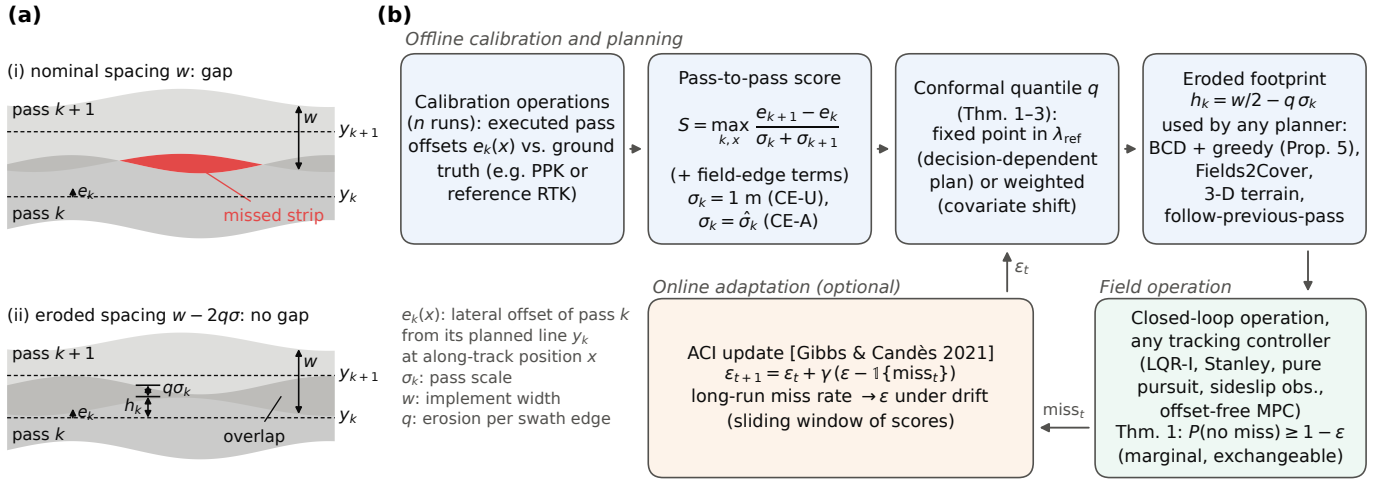

(a)
(i) nominal spacing w: gap
pass $k+1$
w
y_{k+1}
e_k
y_k
pass k
missed strip
(ii) eroded spacing $w - 2q\sigma$: no gap
pass $k+1$
w
y_{k+1}
$q\sigma_k$
h_k
e_k
y_k
pass k
overlap
$e_k(x)$: lateral offset of pass k from its planned line y_k at along-track position x
σ_k: pass scale
w: implement width
q: erosion per swath edge
(b)
Offline calibration and planning
Calibration operations (n runs): executed pass offsets $e_k(x)$ vs. ground truth (e.g. PPK or reference RTK)
Pass-to-pass score
$S = \max_{k,x} \frac{e_{k+1} - e_k}{\sigma_k + \sigma_{k+1}}$
(+ field-edge terms)
$\sigma_k = 1$ m (CE-U),
$\sigma_k = \hat{\sigma}_k$ (CE-A)
Conformal quantile q
(Thm. 1–3):
fixed point in λ_{ref}
(decision-dependent plan) or weighted (covariate shift)
Eroded footprint
$h_k = w/2 - q\,\sigma_k$
used by any planner:
BCD + greedy (Prop. 5), Fields2Cover, 3-D terrain, follow-previous-pass
ε_t
Online adaptation (optional)
ACI update [Gibbs & Candès 2021]
$\varepsilon_{t+1} = \varepsilon_t + \gamma(\varepsilon - \mathbb{1}\{\text{miss}_t\})$
long-run miss rate $\to \varepsilon$ under drift
(sliding window of scores)
miss_t
Field operation
Closed-loop operation, any tracking controller (LQR-I, Stanley, pure pursuit, sideslip obs., offset-free MPC)
Thm. 1: $P(\text{no miss}) \geq 1 - \varepsilon$
(marginal, exchangeable)